Question Paper Preview

Notations :

1.Options shown in green color and with ✔ icon are correct.

2.Options shown in red color and with ✖ icon are incorrect.

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Corporate Finance

Section Id :	640653124559
Number of Questions :	31
Section Marks :	100
Maximum Instruction Time :	0

Question Number : 1 Question Id : 6406531859592 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : CORPORATE FINANCE (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536025427. ✔ YES

6406536025428. ✖ NO

Question Number : 2 Question Id : 6406531859593 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

A firm is considering an investment opportunity promising to pay Rs 8,000 in one year and Rs 12,000 in two years. If the prevailing constant rate of return is 6%, what is the maximum price the firm should pay for the project? (Round off to the nearest integer)

Options :

6406536025429. ✖ Rs 12,000

6406536025430. ✖ Rs 16,350

6406536025431. ✔ Rs 18,227

6406536025432. ✖ Rs 20,000

Question Number : 3 Question Id : 6406531859594 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

A stock currently trading at Rs 60 can either rise to Rs 75 or fall to Rs 50 in three months. The three-month risk-free rate is 5%. Using the risk-neutral approach, what is the risk-neutral probability that the stock price increases?

Options :

6406536025433. ✔ 52.0%

6406536025434. ✖ 78.6%

6406536025435. ✖ 60.0%

6406536025436. ✖ 45.0%

Question Number : 4 Question Id : 6406531859595 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

An investor opts for a straddle strategy for his portfolio by purchasing a call option and a put option with strike price of Rs 30 and expiration time of 3 months. The call option price is Rs 3 and put option price is Rs 4. What is maximum loss the investor can incur with this strategy?

Options :

6406536025437. ✔ Rs 7

6406536025438. ✖ Rs 3

6406536025439. ✖ Rs 4

6406536025440. ✖ Rs 1

Question Number : 5 Question Id : 6406531859596 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
If the interest rate in home country (i_t) is 3% and the interest rate in the foreign country (i_t^*) is 6%, what is the home country's expected rate of appreciation of the exchange rate $(\frac{E_{t+1}^e - E_t}{E_t})$ according to the uncovered interest parity condition?
Options :
6406536025441. ✔ 2.91%
6406536025442. ✖ 0.91%
6406536025443. ✖ 5.94%
6406536025444. ✖ 7.94%

Question Number : 6 Question Id : 6406531859597 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
A risky asset has an expected rate of return of 15% and a standard deviation of the rate of return of 20%. If the Sharpe ratio of this risky asset is 0.6, then what is the prevailing risk-free rate of return?
Options :
6406536025445. ✖ 1.00%
6406536025446. ✔ 3.00%
6406536025447. ✖ 4.00%
6406536025448. ✖ 5.00%

Question Number : 7 Question Id : 6406531859598 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
In the Black-Scholes pricing formula for a call option, the Delta (Hedge Ratio) is given by $N(d_1)$. If the Delta for a call option is 0.7, and the current stock price is Rs 100, how many whole shares of the stock are needed to perfectly hedge 10 written call options against a small change in the stock price?
Options :
6406536025449. ✖ 3 shares
6406536025450. ✔ 7 shares
6406536025451. ✖ 10 shares
6406536025452. ✖ Insufficient Information

Question Number : 8 Question Id : 6406531859599 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

A firm is financed with 60% debt and 40% equity. The equity beta is 2.5. The debt has a beta of 0.2. What is the asset beta of the firm?

Options :

6406536025453. ✖ 0.88

6406536025454. ✖ 1.00

6406536025455. ✔ 1.12

6406536025456. ✖ 1.20

Question Number : 9 Question Id : 6406531859600 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Your portfolio consists of two stocks, X and Y, with returns that have a correlation coefficient of 0.5. Stock X has an expected return of 12% and a standard deviation of 20% and Stock Y has an expected return of 10% and a standard deviation of 25%. What is the covariance between the returns of the two stocks?

Options :

6406536025457. ✖ 0.006

6406536025458. ✖ 0.012

6406536025459. ✖ 0.018

6406536025460. ✔ 0.025

Question Number : 10 Question Id : 6406531859601 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

You invest 80% of your money in a security with a beta of 1.5, and the rest of your money in a risk-free security. What is the beta of your resulting portfolio?

Options :

6406536025461. ✖ 0.80

6406536025462. ✔ 1.20

6406536025463. ✖ 1.50

6406536025464. ✖ Insufficient information

Question Number : 11 Question Id : 6406531859602 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

The current market price of a stock is Rs 400. Next year's expected dividend is Rs 25 per share. The dividend growth rate is expected to be 4% per year forever, and the required rate of return is 8%. According to the Gordon Growth Model, what is the estimated fair price of the stock?

Options :

6406536025465. ✖ Rs 312.5

6406536025466. ✖ Rs 400

6406536025467. ✖ Rs 500

6406536025468. ✔ Rs 625

Question Number : 12 Question Id : 6406531859603 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

An investment is expected to yield a total holding period return of 140% over a period of 30 years. What is the annualized rate of return (in percent) on this investment?

Options :

6406536025469. ✖ 1.13%

6406536025470. ✔ 2.96%

6406536025471. ✖ 4.67%

6406536025472. ✖ 1.40%

Question Number : 13 Question Id : 6406531859604 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

A European put option with a strike price of Rs 90 is trading at a price of Rs 5. The stock price is Rs 88, and the risk-free rate is 4% (continuously compounded) for the 1-year time to expiration. According to the put-call parity theorem, what is the corresponding European call option price (in Rupees)?

Options :

6406536025473. ✖ Rs 10.45

6406536025474. ✔ Rs 6.53

6406536025475. ✖ Rs 1.33

6406536025476. ✖ Rs 0.67

Question Number : 14 Question Id : 6406531859605 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

The expected return on the market portfolio is 10% and the risk-free rate is 5%. A security has an expected rate of return of 12.5%. According to CAPM, what is the beta of this security?

Options :

6406536025477. ✖ 0.75

6406536025478. ✖ 1.00

6406536025479. ✔ 1.50

6406536025480. ✖ 2.00

Question Number : 15 Question Id : 6406531859606 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Your expected return on a stock is 15%. The stock has a beta of 1.5. If the risk-free rate is 4% and the expected market return is 10%, what is the stock's alpha (in percent)?

Options :

6406536025481. ✖ -2.0%

6406536025482. ✖ -1.0%

6406536025483. ✖ 1.0%

6406536025484. ✔ 2.0%

Question Number : 16 Question Id : 6406531859607 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

The real interest rate in the US is 4%, and the expected US inflation rate is 1%. What is the nominal interest rate in the US (in percent)?

Options :

6406536025485. ✖ 3.04%

6406536025486. ✖ 4.04%

6406536025487. ✔ 5.04%

6406536025488. ✖ 6.04%

Question Number : 17 Question Id : 6406531859608 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

A long straddle strategy is executed by buying a one-year call option (price Rs 5) and a one year put option (price Rs 3) with the same strike price of Rs 70. If the stock price at expiration is Rs 80, what is the net profit (in Rupees) from this position?

Options :

6406536025489. ✖ Rs 10

6406536025490. ✖ Rs 5

6406536025491. ✔ Rs 2

6406536025492. ✖ There is a loss

Question Number : 18 Question Id : 6406531859609 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

A trader executes a strategy by selling a call option on a stock and simultaneously holding a long position (buying) in the same stock. This strategy is known as a

Options :

6406536025493. ✖ Protective Put

6406536025494. ✖ Long Straddle

6406536025495. ✔ Covered Call

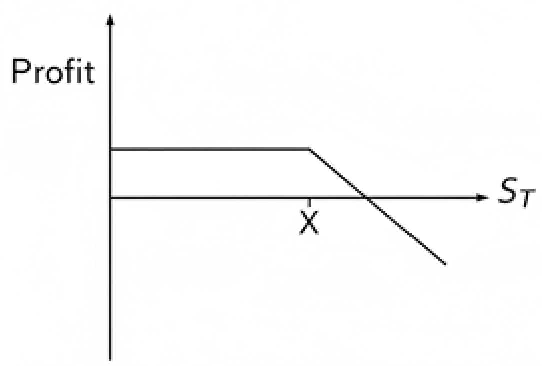
6406536025496. ✖ Long Call

Question Number : 19 Question Id : 6406531859610 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

What option position results in the profit diagram as given in the figure, on the exercise date? S_T denotes the price of the underlying asset and X denotes the strike price.



Options :

6406536025497. ✖ Long call

6406536025498. ✔ Short call

6406536025499. ✖ Long put

6406536025500. ✖ Short put

Question Number : 20 Question Id : 6406531859611 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Kartik’s utility function can be written as $U(W) = \sqrt{(2W)}$, where W is his wealth level. He currently has wealth equal to 200 thousand dollars. He invests his wealth in an asset, which will give a return of 5% with probability 0.75, or a return of -5% (that is, a return of negative 5%) with remaining probability by the end of the year (His wealth will not change due to any other reason). This asset gives him as much expected utility as if he invested his wealth in risk-free bonds with an interest rate of x% per year. What is the value of x?

Options :

6406536025501. ✖ -1.45%

6406536025502. ✖ 0.00%

6406536025503. ✔ 2.45%

6406536025504. ✖ 4.34%

Question Number : 21 Question Id : 6406531859612 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider a stock whose return over the next year is sensitive to one factor – return on an index fund (r_m). Suppose that the sensitivity of the stock return to the index fund is 1.2. If the risk-free interest rate is 5% and risk premium on index fund is 3%, then what is the expected return on the stock over the next year, according to the Arbitrage Pricing Theory?

Options :

6406536025505. ✖ 5.0%

6406536025506. ✖ 2.6%

6406536025507. ✔ 8.6%

6406536025508. ✖ 3.6%

Question Number : 22 Question Id : 6406531859613 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

A risky asset has an expected rate of return of 16% and a standard deviation of 20%. If the risk-free rate is 4%, select the correct statements.

Options :

6406536025509. ✔ The Sharpe ratio of the asset is 0.6.

6406536025510. ✔ The risk premium of the asset is 12%.

6406536025511. ✖ The slope of the capital allocation line passing through this asset is 0.8.

6406536025512. ✔ If an investor with utility function $U = \mu - 2\sigma^2$ combines this asset with the risk-free asset, the optimal weight allocated to the risky asset is 0.75.

Question Number : 23 Question Id : 6406531859614 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a put option and a call option on the same stock with the same strike price of Rs 250 and time to maturity of three months. Select the correct statements

Options :

6406536025513. ✔ If the stock price at expiration is Rs 260, call option is in the money.

6406536025514. ✔ If the stock price at expiration is Rs 260, put option is out of the money.

6406536025515. ✖ If the stock price at expiration is Rs 240, call option is in the money.

6406536025516. ✔ If the stock price at expiration is Rs 240, put option is in the money.

Question Number : 24 Question Id : 6406531859615 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

A trader buys a call option on a stock with a strike price of Rs 40 when the call option price is Rs 4. He also buys two put options with the same strike price of Rs 40 and the put option price is Rs 3. The trader makes a profit when the stock price at expiration is

Options :

6406536025517. ✔ Rs 32

6406536025518. ✖ Rs 40

6406536025519. ✖ Rs 48

6406536025520. ✔ Rs 56

Question Number : 25 Question Id : 6406531859616 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

A call option on a non-dividend-paying stock has a hedge ratio of 0.6, select the correct statements

Options :

6406536025521. ✔ If the stock price increases by Rs 1, the call option price increases by approximately Rs 0.6.

6406536025522. ✔ If an investor sells 10 call options, she must buy 6 shares of the underlying stock to form a delta-hedged position.

6406536025523. ✔ A put option with same strike price and expiry time has Delta of -0.4.

6406536025524. ✖ The put option price increases if the stock price increases, as Delta is positive.

Question Number : 26 Question Id : 6406531859617 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

The price of a one-year European call option on a non-dividend-paying stock is Rs 15 (C_0) which has a strike price of Rs 120 (X). The stock price is Rs 115 (S_0), and the annual risk-free rate is 8% (continuously compounded, r). Select the correct statements.

Options :

6406536025525. ✖ A one-year call option with strike price of Rs 125 should have price more than Rs 15.

6406536025526. ✔ A six-month call option with strike price of Rs 120 should have price less than Rs 15.

6406536025527. ✔ The price of a one-year European put option on the same stock is approximately Rs 10.77.

6406536025528. ✖ The sum $C_0 + S_0$ is equal to $P_0 + Xe^{-rT}$.

Question Number : 27 Question Id : 6406531859618 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question
Security A has an expected return of 14% and a standard deviation of 20%. Security B has an expected return of 18% and a standard deviation of 25%. Suppose that the rates of return of the two securities have a correlation coefficient of 0.8. Based on the given information select the correct statements from below.
Options :
6406536025529. ✖ The covariance of security A and security B is 1.5%.
6406536025530. ✔ If an investor invests 50% of wealth in each security, then his portfolio's expected return is 16%.
6406536025531. ✔ If an investor invests 50% of wealth in each security, then his portfolio's standard deviation is 21.36%.
6406536025532. ✔ If short selling is not allowed, it is not possible to build a portfolio with expected return of 20%.

Question Number : 28 Question Id : 6406531859619 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question
A money manager calculates that JLK Inc stock has an expected rate of return of 16%. JLK Inc has a beta of 1.5. The risk-free rate of return is 6%, and the expected market rate of return is 12%. Which of the following statements are correct?
Options :
6406536025533. ✖ JLK stock is overpriced.
6406536025534. ✔ JLK stock is underpriced.
6406536025535. ✔ JLK stock would lie above the security market line.
6406536025536. ✖ JLK stock would lie below the security market line.

Question Number : 29 Question Id : 6406531859620 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question
A stock is trading at Rs 100 today. In 6 months, the stock price can either increase to Rs 120 or decrease to Rs 90. The six-month risk-free rate is 2%. A six-month call option has an exercise price of Rs 110. Select the correct statements based on the given information.
Options :
6406536025537. ✔ The risk-neutral probability that stock price increases is 40%.
6406536025538. ✖ The hedge ratio for the given call option is 2/3.
6406536025539. ✔ The purchase of one share can be hedged with 3 call options.
6406536025540. ✔ According to risk-neutral approach, the price of call option is Rs 3.92.

Question Number : 30 Question Id : 6406531859621 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question
KLM Inc. has an expected return of 12% and a beta of 1.6. The risk-free rate is 6% and the expected return on market portfolio is 10%. Select the correct statements according to CAPM
Options :
6406536025541. ✔ The required expected return for KLM Inc. should be 12.4%.
6406536025542. ✔ KLM Inc. stock is overpriced.
6406536025543. ✖ KLM Inc. has a positive Alpha of 0.4%.
6406536025544. ✔ An investor following the CAPM recommendation should sell KLM Inc. shares.

Question Number : 31 Question Id : 6406531859622 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question
An investor is considering two assets for her portfolio: a risk-free asset with expected return of 5% and a risky asset with expected return of 12% and standard deviation of 10%. Her utility function is given as $U = \mu - 5\sigma^2$. Given this information select the correct statements.
Options :
6406536025545. ✔ The optimal weight allocated to the risky asset for the investor is 70%.
6406536025546. ✔ The standard deviation of the optimal portfolio is 7%.
6406536025547. ✔ The expected return of the optimal portfolio is 9.9%.
6406536025548. ✔ The Sharpe ratio of the risky asset is 0.7.